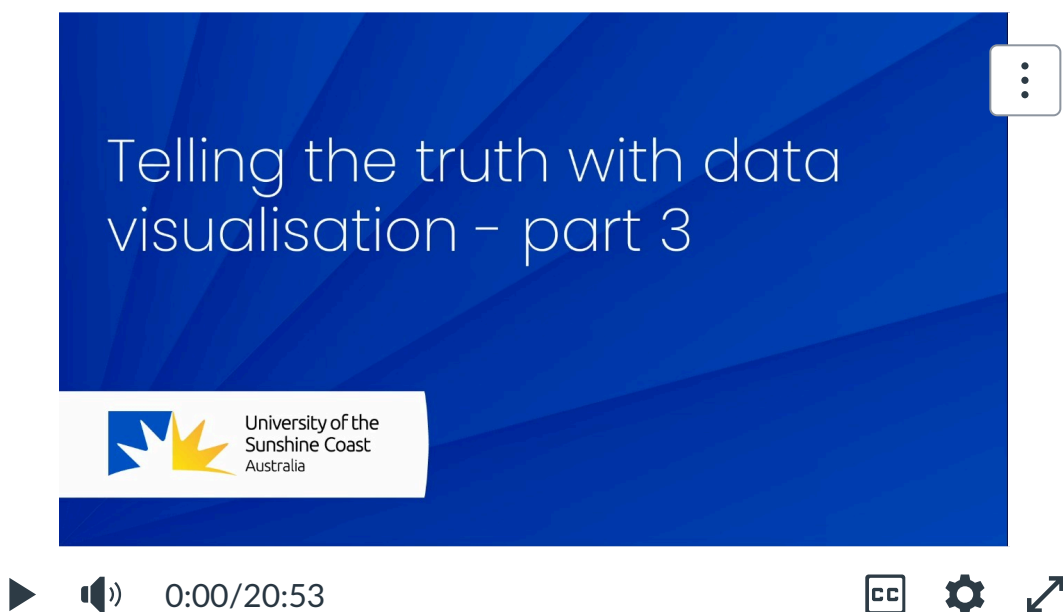


10.4 Deceptive design ↕

- Watch: Telling the truth with data visualisation (part 3)
[20:53 mins]

Watch the video below, which explains different ways in which data can be misrepresented through inappropriate chart design.

This is useful as it provides awareness of improper chart choices that can lead to deceptive data visualisations.



Transcript

(<https://learn.usc.edu.au/courses/35242/files/3138652?wrap=1>).

Source: © University of the Sunshine Coast (2023).

Download the following slides for note taking: [ICT702 Telling the truth with data visualisation \(part 3\)](https://learn.usc.edu.au/courses/35242/files/3168555?wrap=1).(<https://learn.usc.edu.au/courses/35242/files/3168555?wrap=1>)

The goal of a visualisation is to clearly and accurately convey insights from the data to an audience. Therefore, deceptive designs that result in misleading the audience must be avoided. A data visualisation designer must be very careful when making decisions about each visualisation. Various issues must be considered.

Design of chart axes

What you choose for the axes of a chart may impact the insights perceived by an audience.

Problems may occur when:

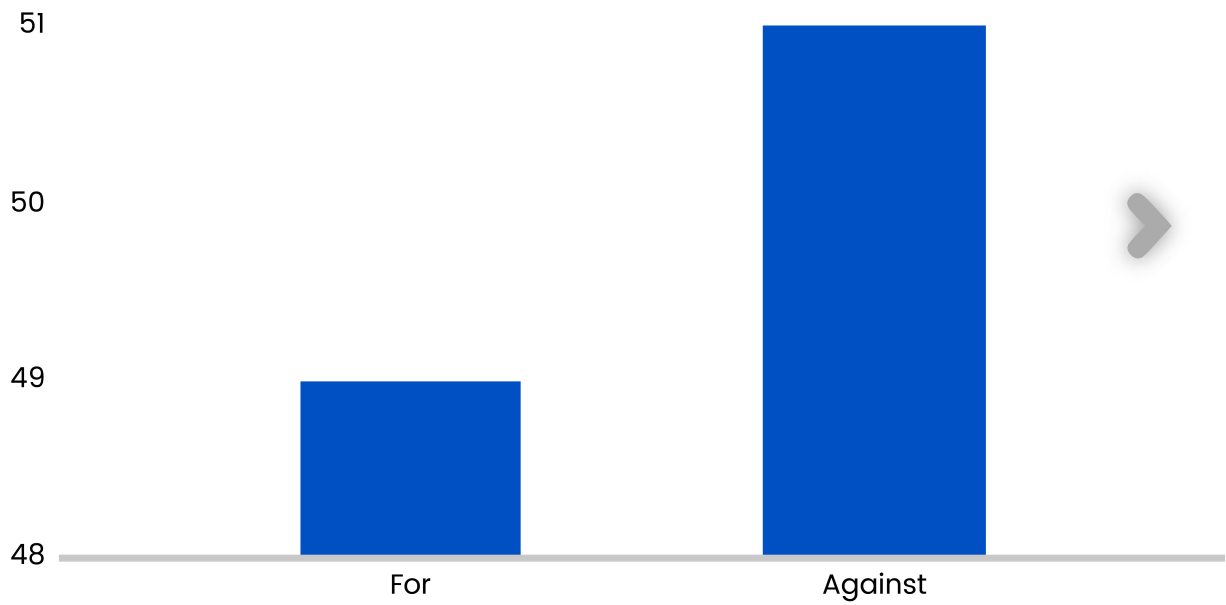
- *changing the lower and/or upper limits of an axis (horizontal or vertical)*. In general, column and bar chart axes should always start at zero; exceptions include, but are not limited to, situations where negative and positive values need to be shown on the same chart. When charts use orientation or spatial positioning (e.g. line charts and scatter charts), it may be better to use a smaller range on the vertical axis to enable the variability, trend or correlation in the data to be shown.
- *changing the aspect ratio of a chart*. Changing the horizontal or vertical size of a chart may mislead an audience. Tall, narrow charts have a smaller aspect ratio and tend to exaggerate trends in the data. Short, wide charts have a larger aspect ratio and tend to disguise trends in data.

□ Activity: Compare the graphs 2

Compare the two graphs below and identify the differences. Click on the solution to check your understanding.

Majority of voters against library levy

Percent (%) of likely voters



Source: Camm et al. (2021), pp. 377–78.

> **Solution**

Dual-axis charts

It may be necessary to show an audience two different variables from a dataset. Each variable may have different units and/or different magnitudes. It is possible to achieve this on one chart: a dual-axis chart. On this chart, a secondary axis is used to represent one of the variables. However, the result is often difficult to interpret, and incorrect conclusions may be drawn as inaccurate comparisons may be made.

A better solution is to create two charts with a single variable shown on each. Using this option means there will be less likelihood of the audience making comparisons.

□ Activity: How EV adoption will impact oil consumption (2015–2025P)

Reflect on this webpage about how EV adoption will impact oil consumption. Note the webpage was published in May 2023.

As you look at the visualisation, consider and make notes on:

- the “y” axes. Both represent barrels per day – might audiences be misled; if so, why?
- how time is represented.
- how you might visualise the data.
- what techniques you would use, and why.

Explain your reasoning and bring your notes to the workshop.

Source: Oğuz, S. (2023, May 15). [How EV adoption will impact oil consumption \(2015-2025P\)](https://www.visualcapitalist.com/how-ev-adoption-will-impact-oil-consumption-2015-2025p/). *Visual Capitalist*.

Data selection and temporal frequency

The insights conveyed to an audience are directly related to the choice of data included in a chart. While this may be an obvious fact, it is particularly important to take this into consideration with time series data – i.e. data that is collected over successive points in time. The temporal frequency of the plotted value points in a chart can convey misleading insights to an audience. Plotting charts using a lower temporal frequency produces a smoothing effect (reducing apparent variability), which may hide seasonal or cyclic variations present in the data.

It is often useful to view time series data at different temporal frequencies to reveal patterns that exist at one frequency but are hidden at another. In general, it is useful to collect data at as frequent time intervals as possible as it allows for more variation of the temporal frequencies that can be visualised.

Issues related to geographic maps

Geographic maps are useful for visualising various data, including quantitative variables. However, choropleth maps – which use different shades of colour to represent different quantitative values from a range of values – can be misleading if other environmental factors are not taken into consideration (e.g. a country's population varies across areas within that country). This can distort the representation of another variable, such as the rate of poverty in each area. In general, when creating geographic maps, it is better to use the relative values of quantitative variables.